

CS202: Advanced Algorithms & Data Structures Syllabus

Unit 1: Linear & Linked Structures

Topic: Linked Lists - Singly Linked Lists, Doubly Linked Lists

Topic: Stacks and Queues - LIFO Mechanics, Priority Queues

Unit 2: Non-Linear Hierarchies

Topic: Tree Traversal - Binary Search Trees, Balanced AVL Trees

Topic: Heap Structures - Min Heaps, Heap Sort

Unit 3: Graph Paradigms

Topic: Graph Representations - Adjacency Matrices, BFS & DFS

Topic: Shortest Paths - Dijkstra Algorithm, Bellman-Ford